Maxwell Tsai

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EDUCATION

Boston University – College of Engineering Expected May 2027
Bachelor of Science in Electrical Engineering, Concentration in Robotics GPA: 3.96/4.00

Coursework: Robotics, Control Systems, Power Electronics (Spring 2027), Digital Image Processing (Spring 2027), Engineering Mechanics, Signals and Systems, Probability and Statistics, Software Engineering

PROJECTS

BusBot: Autonomous Dish-Collection Robot *(Senior Design)* May 2026 – Present

- Served as initial team lead of a 6-person team (2 EE, 2 CE, 2 ME), defining system architecture (ESP32-S3 sensor controller, Jetson Orin Nano running ROS 2 for LiDAR navigation) and requirements from a 25-diner survey
- Defined sensor coverage for LiDAR navigation, adding rear ultrasonic sensing to cover the 84° blind spot of the 276° 2D LiDAR out to a 20 cm obstacle buffer
- Wrote ESP32-S3 firmware (ESP-IDF, C) for obstacle stopping with PWM motor control and quadrature encoder feedback: brakes at 40 cm, resumes past 45 cm (5 cm hysteresis), and stops on sensor failure
- Validated on a motor bench rig: ±2 cm range accuracy, brake command under 1 ms after detection, 5/5 stop/restart trials
- Built component-level power budget (~44 W avg) to size a 12 V 18 Ah SLA battery for 2.5 hr at 50% depth of discharge

Autonomous Laundry Folder Jan 2026 – May 2026

- Integrated power and control for a SCARA-inspired 2R1P manipulator (laser-cut parallel-plank linkage, aluminum extrusion frame) as the sole EE on a 5-person team
- Derived forward kinematics by hand and co-planned a 12-step folding trajectory across a 70×60×20 cm workspace
- Debugged Arduino joint control against a teammate's inverse kinematics solution, reaching correct joint angles at each step
- Powered two 25 kg·cm linkage servos and an electromagnet end effector from a 12 V supply through buck converters

Bedside Beacon: Point-to-Point Communication Device Sep 2025 – Dec 2025

- Wired 90° servo-steered laser transmitter and designed and built laser-cut housing in Onshape; 19 of 21 transmissions succeeded from 4 to 8 ft against a 5 ft spec

EXPERIENCE

Intel Hillsboro, OR
Data Science and Analytics Intern May 2026 – Aug 2026

- Traced ridge regression errors predicting in-line electrical test values to outdated metrology revisions in training data
- Built a JMP Scripting Language tool that keeps latest revision per measurement per wafer, cutting data retrieval from 2+ weeks to 2 hours (~100 measurements, ~200 wafers); still used by device engineers after internship

Analog Product Development & Test Engineer Intern May 2025 – Aug 2025

- Designed an object-oriented Python API abstracting 4 lab instruments as generalized scopes, clocks, and sources, letting engineers test remotely with common functions instead of instrument-specific commands
- Automated IEEE 488.2 SCPI communication over GPIB (PyVISA) to configure instruments and transfer measurement data
- Wrote ~100 pytest unit tests covering API functions for each instrument and refactored 7,000+ lines of Python to PEP 8

Boston University Dept. of Electrical and Computer Engineering Boston, MA
Teaching Assistant – Logic Design Jan 2026 – May 2026

- Led Verilog FPGA lab sections in Vivado and debugged student designs; graded homework and exams for 80+ students

SemeterLab at Boston University Boston, MA
Undergraduate Research Assistant Sep 2025 – Dec 2025

- Developed an Android app logging Global Navigation Satellite System (GNSS) measurements for real-time ionospheric monitoring, tracking 15+ satellites and exporting carrier-to-noise density (C/N0) data

SKILLS

Robotics: robot kinematics, trajectory planning, motor control (PWM, encoders), sensor integration (ultrasonic, voltage/current sense)
Embedded: ESP32-S3 (ESP-IDF), Arduino, I2C
Programming: Python, C, C++, MATLAB
Tools: Git, Linux, Onshape (CAD), laser cutting, soldering

LEADERSHIP

Theta Tau Professional Engineering Fraternity (New Member Education, Professional Development, and Brotherhood Chair): ran resume and interview workshops and coached 20+ members 1-on-1
BostonU Club Tennis (Community Service Chair): organized youth field day service event